InterviewAI: AI-Powered Adaptive Interview Preparation and Evaluation System

Technical interviews are one of the most important stages in hiring for roles such as software engineers, data scientists, and analysts. However, preparing effectively for these interviews is challenging for candidates. Most existing preparation platforms provide generic question banks, static quizzes, or isolated coding problems, which do not adapt to a single question-answer interaction and lack realistic interviewer behavior.an individual candidate's knowledge level, strengths, or weaknesses.

At the same time, real-world interviews conducted by companies involve multiple interviewers evaluating different dimensions, such as technical knowledge, system design skills, reasoning ability, and communication. Traditional preparation tools rarely replicate this multi-interviewer environment or provide detailed feedback on candidate performance.

As a result, candidates often practice in an unstructured manner, spending time on topics they already know while overlooking critical conceptual gaps.

Current interview preparation tools suffer from several key limitations:

1. **Lack of Personalization**
   Most platforms provide standardized question sets that are not tailored to a candidate's background, target role, or experience level.

2. **No Conceptual Understanding of Skills**
   Existing systems treat topics independently and fail to capture relationships between concepts. For example, understanding Random Forest requires knowledge of statistics, decision trees, and bias–variance tradeoff. Without mapping these dependencies, tools cannot diagnose root learning gaps.

3. **Limited Interview Simulation**
   Real interviews involve multiple perspectives (technical, system design, behavioral). However, most preparation platforms simulate on

4. **Weak Feedback Mechanisms**
   Candidates often receive only basic correctness feedback. They rarely receive detailed insights into:
   - conceptual gaps
   - reasoning quality
   - communication clarity
   - behavioral signals

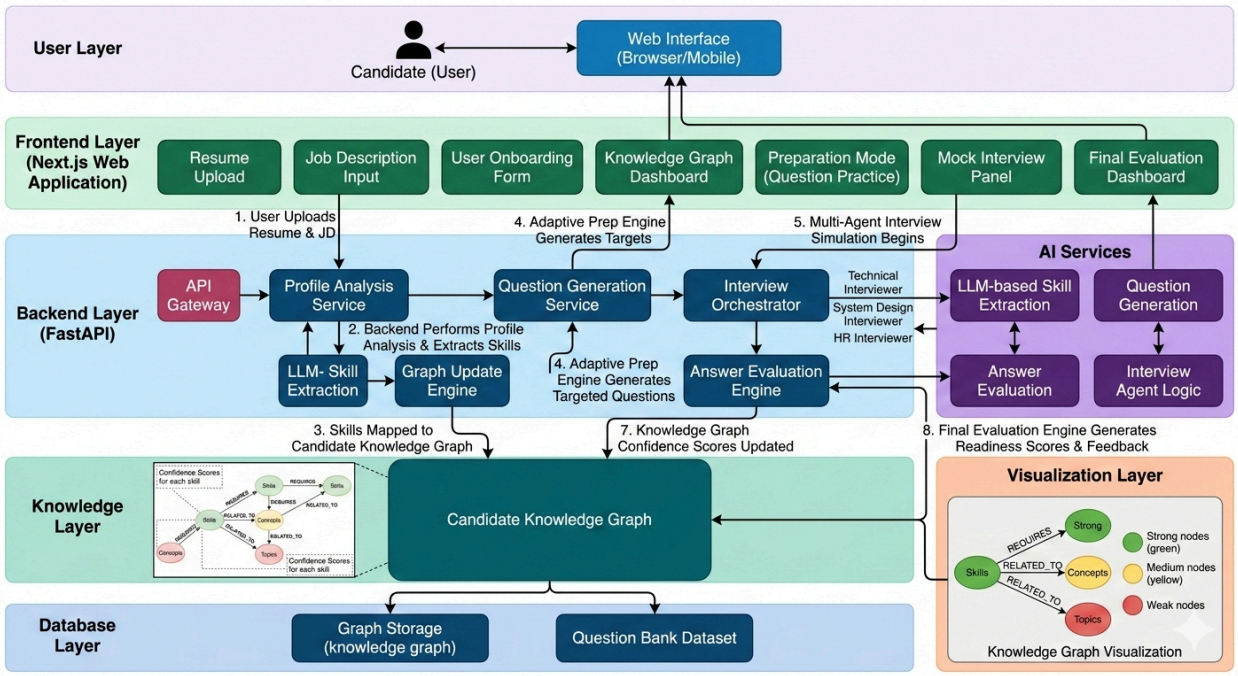
5. **No Adaptive Learning Loop**
   Effective preparation should dynamically adapt based on a candidate's responses. Existing systems do not adjust question difficulty or topic focus in real time.

To address these issues, there is a need for an intelligent system that can:

1. Analyze a candidate's resume, job description, and experience level
2. Build a structured representation of the candidate's knowledge

3. Identify concept dependencies and weak areas
4. Provide adaptive questions targeting those weaknesses
5. Simulate real interview environments with multiple interviewers
6. Generate detailed evaluation and learning feedback

Solution Architecture



1. The Intake & Analysis Phase

Everything starts with data ingestion. This layer translates raw documents into a structured "Skill Map."

The Action: You upload your Resume and a Job Description (JD) via the Next.js frontend.

The Logic: The Profile Analysis Service (FastAPI) sends this text to the LLM Skill Extraction engine.

The Output: Instead of just a list of words, the system identifies specific skills and their relationships. For example, it doesn't just see "Python"; it sees "Python" as a REQUIRED node for "Data Science."

2. The Knowledge Graph (The "Brain")

This is the core differentiator of the platform. Instead of a flat database, it uses a Knowledge Graph.

The Nodes: Represent Skills (e.g., React), Concepts (e.g., Hooks), and Topics (e.g., State Management).

The Edges: Define relationships like REQUIRES or RELATED_TO.

Confidence Scores: Every node starts with a baseline score derived from your resume. As you answer questions, these scores fluctuate.

Visualization: The Visualization Layer renders this graph using a traffic-light system:

1. **Green:** Strong mastery.

2. **Yellow:** Needs a refresher.
3. **Red:** Significant knowledge gap.

3. The Adaptive Preparation Engine
Before the full simulation, the platform helps you patch holes in your knowledge.
Targeted Generation: The Question Generation Service queries the Knowledge Graph for "Red" or "Weak" nodes.
Precision: If your graph shows you know "SQL" but are weak in "Window Functions," the engine generates questions specifically for that sub-topic rather than general database questions.

4. The Multi-Agent Interview Simulation
This is the "live" part of the platform where the Interview Orchestrator manages three distinct AI personas.
Technical Interviewer: Focuses on code, logic, and the "How."
System Design Interviewer: Focuses on scalability, architecture, and the "Why."
HR Interviewer: Focuses on culture fit and behavioral questions (STAR method).
Inter-Agent Logic: These agents don't just read a script; they react to your previous answers, drilling deeper if you seem unsure or skipping ahead if you show mastery.

5. Evaluation & Feedback Loop
The final chunk closes the loop, turning your performance into data.
Answer Evaluation Engine: Uses LLMs to grade your responses based on accuracy, clarity, and depth.
Graph Update Engine: This is crucial. If you nail a difficult system design question, the "Confidence Score" for that node in your Knowledge Graph increases.
Final Dashboard: You receive a Readiness Score and a refreshed visualization of your graph, showing exactly how much "Greener" your profile became after the session.

Technology Stack

1. **Frontend Framework – Next.js**
   Used to build the interactive web application and manage routing, UI rendering, and user workflows.
2. **UI Styling – Tailwind CSS**
   Provides a utility-first styling approach for creating responsive and consistent user interfaces.
3. **UI Components – shadcn UI**
   Supplies pre-built reusable components such as forms, cards, and buttons for faster development.

4. **Graph Visualization – React Flow**
   Used to display the **candidate knowledge graph**, where nodes represent skills/concepts and edges represent dependencies.
5. **Backend Framework – FastAPI**
   Handles REST API development, backend logic, and communication between frontend, AI services, and the knowledge graph.
6. **Programming Language – Python**
   Used for implementing backend services, AI integrations, and graph processing logic.
7. **Artificial Intelligence – Large Language Models (LLMs)**
   Used for skill extraction from resumes/JDs, adaptive question generation, answer evaluation, and multi-agent interview simulation.
8. **Knowledge Graph Engine – NetworkX (Python)**
   Represents candidate skills and concept dependencies as a graph structure with confidence scores.
9. **API Communication – REST APIs**
   Enables data exchange between frontend and backend for profile analysis, question generation, graph retrieval, and evaluation.
10. **Development Tools – Node.js, npm, Git**
    Used for frontend dependency management, development workflow, and version control.

Demo Instructions

To demonstrate the AI-powered interview preparation system, the application must first be set up and executed in a local environment. The system consists of a frontend interface built using Next.js and a backend service developed with FastAPI. Before running the application, ensure that the necessary dependencies such as Node.js, npm, Python (version 3.9 or above), and Git are installed. The backend dependencies can be installed using the project's requirements file, and the frontend dependencies can be installed using npm.

Once the dependencies are installed, the backend server should be started by navigating to the backend directory and running the FastAPI application. This will launch the backend service on a local server, which manages API requests, profile analysis, question generation, and answer evaluation. The API documentation can also be accessed through the automatically generated Swagger interface to verify that all endpoints are functioning correctly.

After starting the backend, the frontend application can be launched by navigating to the frontend directory and running the development server. The web application will then be available in a browser

through the local host address. This interface allows users to interact with the system and access all its functionalities.

To begin the demonstration, the user uploads a resume or enters a job description along with optional details such as the target role and target company. Once the information is submitted, the system analyzes the provided input and extracts relevant skills, technologies, and concepts. This information forms the basis for building the candidate's knowledge representation.

Following this step, the system presents a short onboarding questionnaire to gather additional information about the candidate. These questions may include the user's years of experience, preferred technical domains, technologies they are comfortable with, and areas where they feel less confident. The onboarding responses help the system personalize the interview preparation process.

After the analysis is completed, the system generates a knowledge graph dashboard that visually represents the candidate's skills and conceptual relationships. Each node in the graph corresponds to a specific skill or concept, and the nodes are color-coded based on the system's confidence level in the candidate's knowledge. Strong areas appear in green, moderate areas in yellow, and weak areas in red, allowing users to quickly identify topics that require improvement.

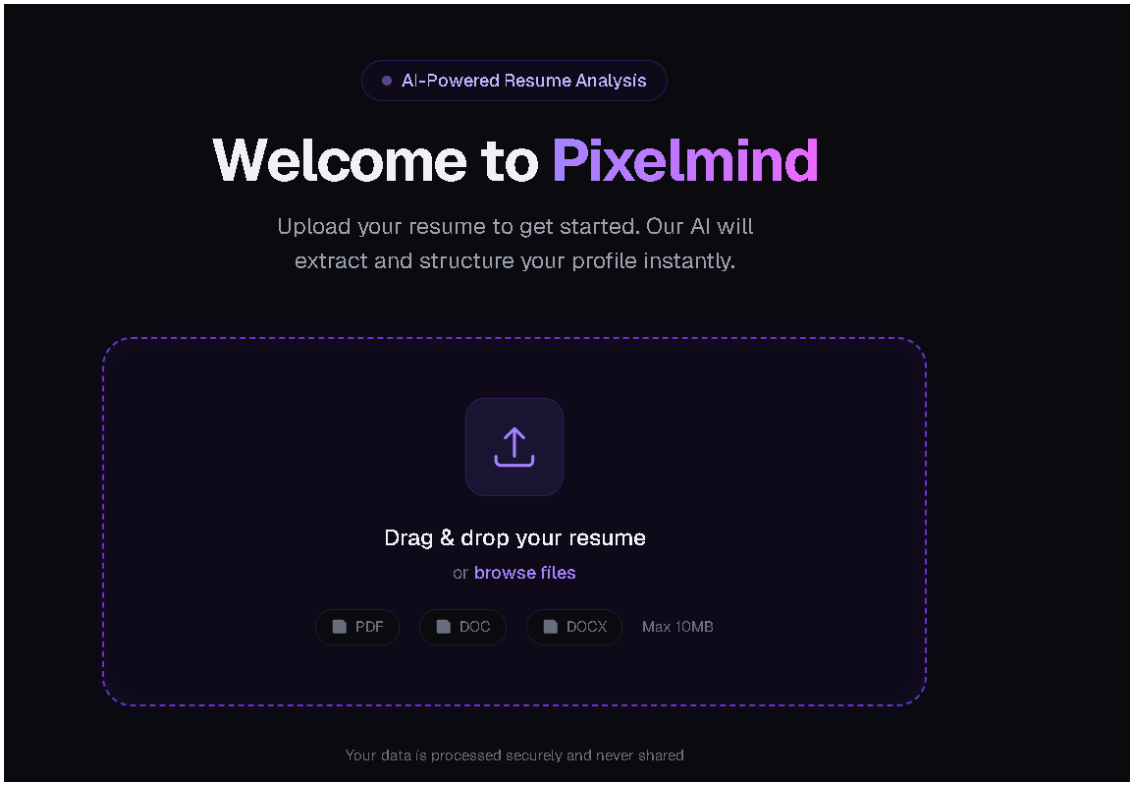
The candidate can then enter the preparation mode, where the system generates adaptive practice questions. These questions are selected based on weaker nodes in the knowledge graph and are designed to improve the candidate's understanding of specific concepts. The system evaluates each answer and dynamically updates the graph based on the candidate's performance.

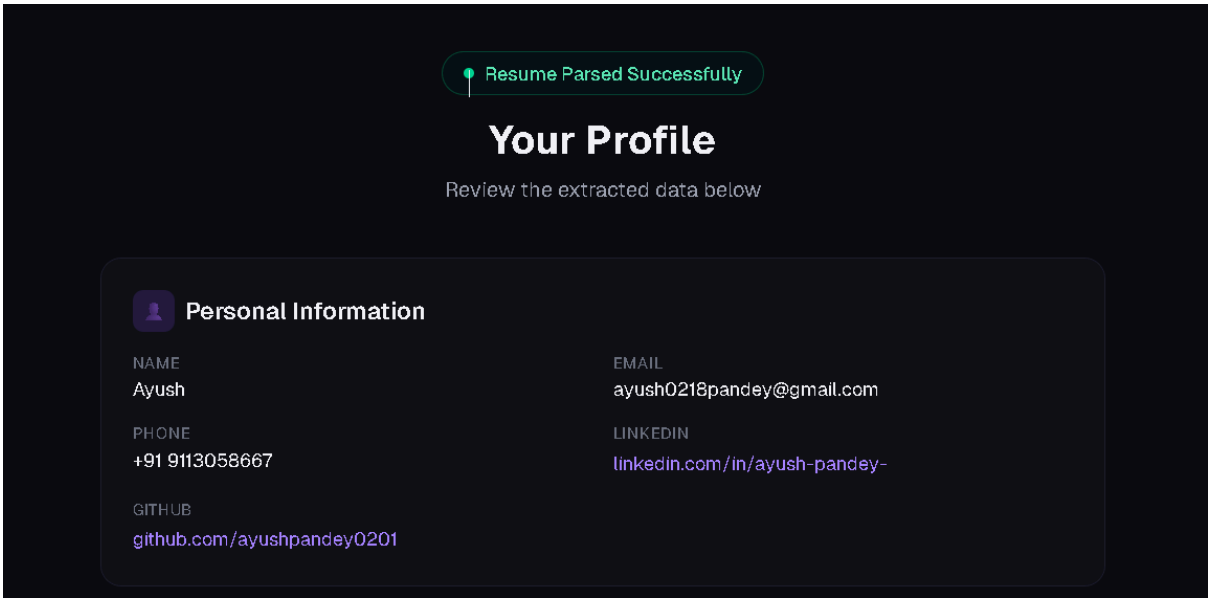
Next, the system allows the user to participate in a mock interview simulation. During this phase, multiple AI-driven interviewer agents simulate a realistic interview panel. These agents represent different roles such as technical interviewer, system design interviewer, and HR interviewer. Each agent asks context-aware questions and may generate follow-up questions depending on the user's responses.

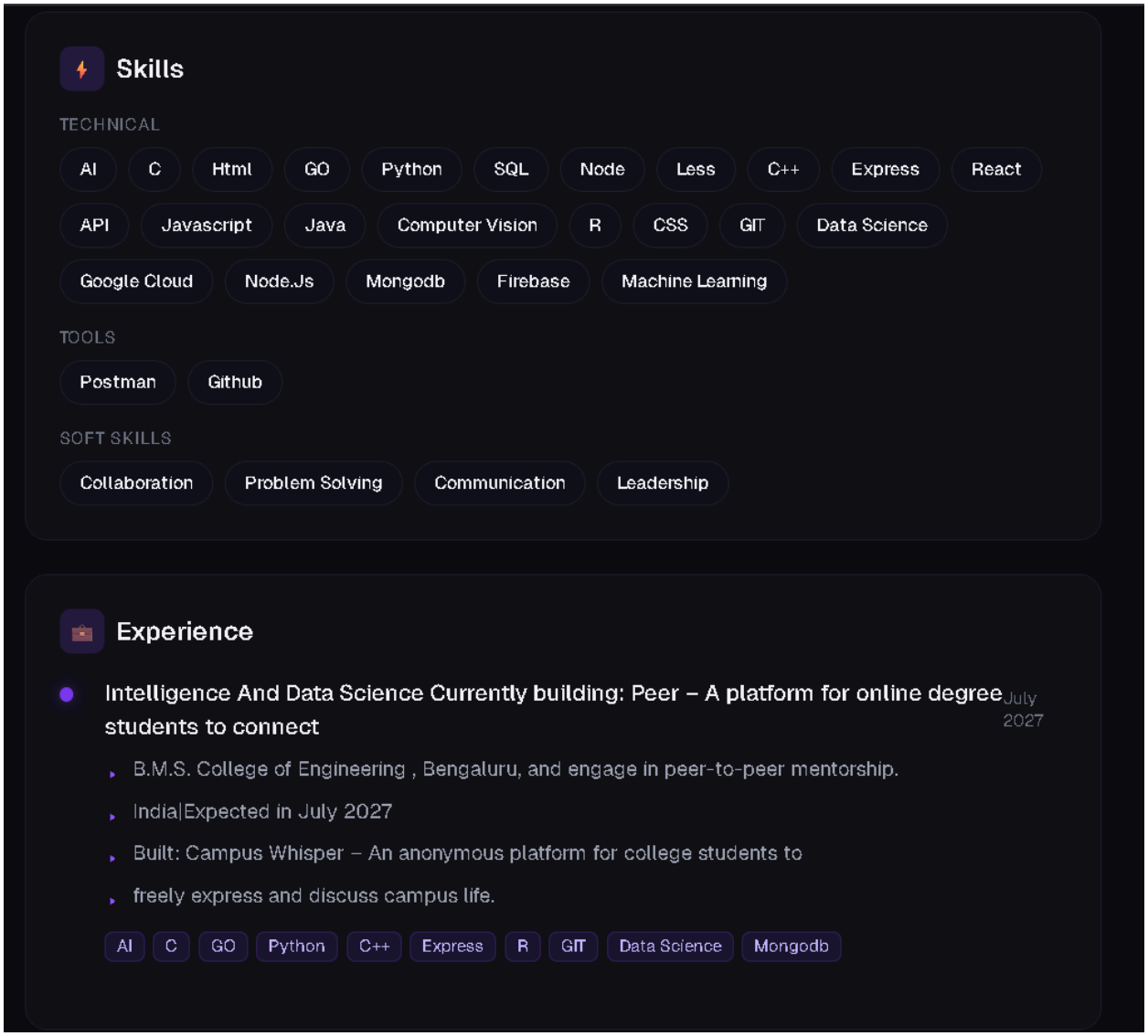
At the end of the interview session, the system provides a comprehensive evaluation dashboard. This dashboard includes scores for different dimensions such as technical knowledge, system design ability, communication clarity, and reasoning depth. The system also calculates an overall interview readiness score based on the candidate's performance.

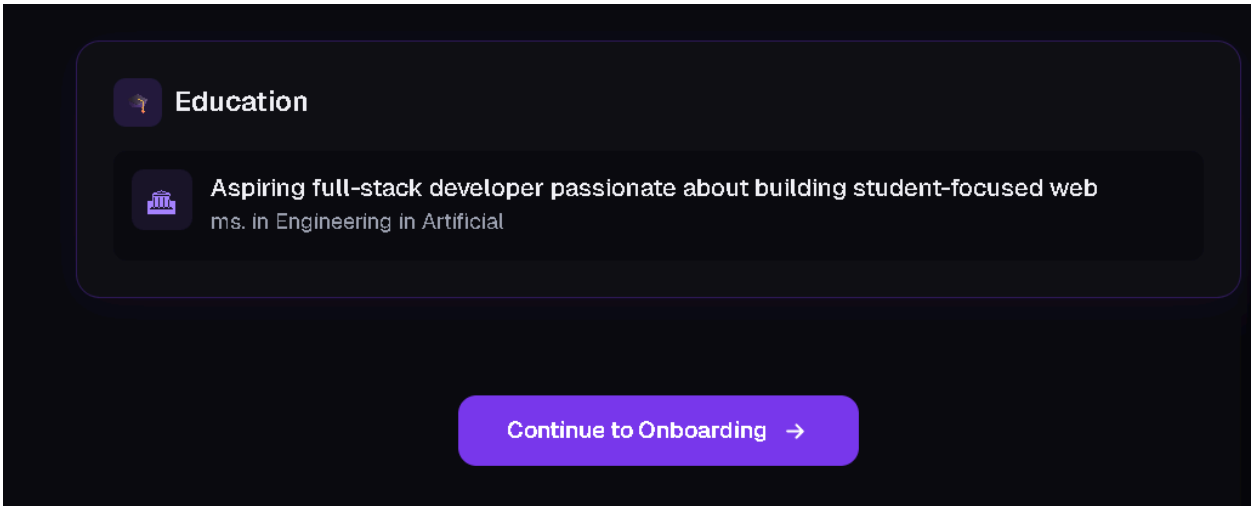
Finally, the system presents visual feedback through the updated knowledge graph, highlighting areas that still require improvement. It also provides recommendations for further preparation, allowing the candidate to focus on strengthening specific skills and concepts before attempting real interviews.

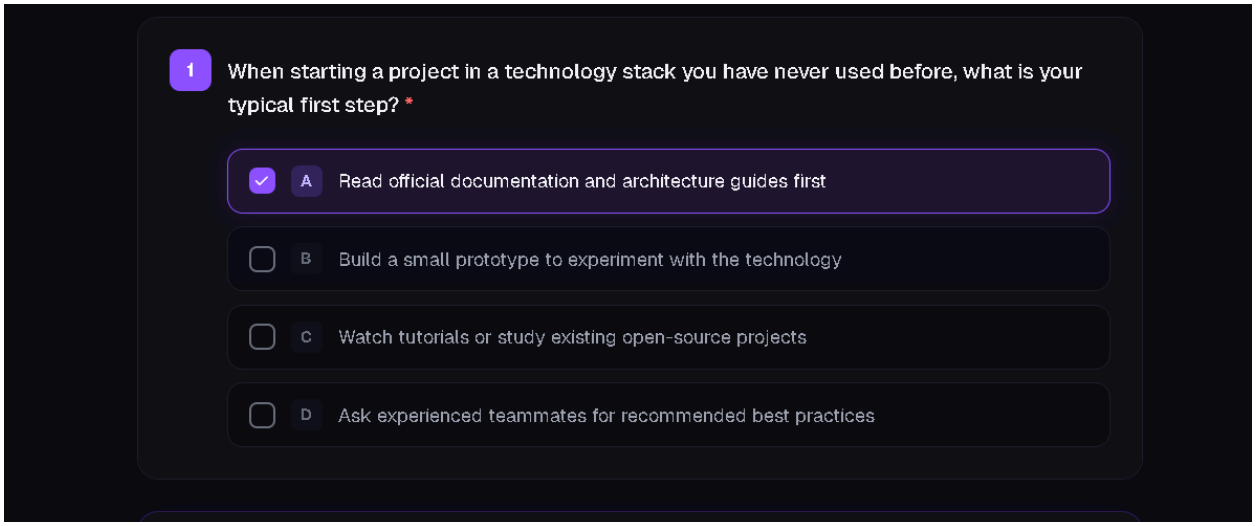
Screenshots

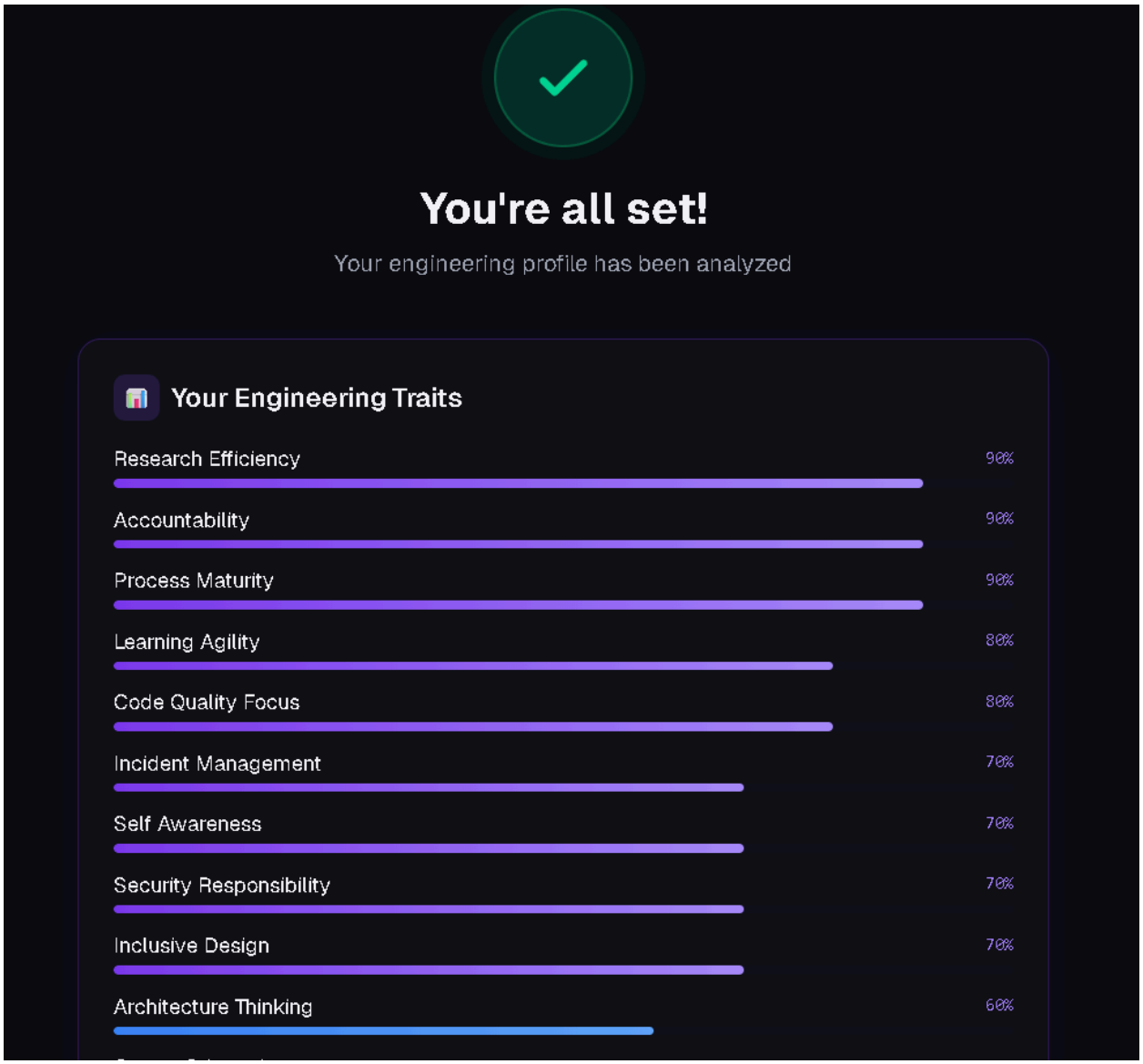


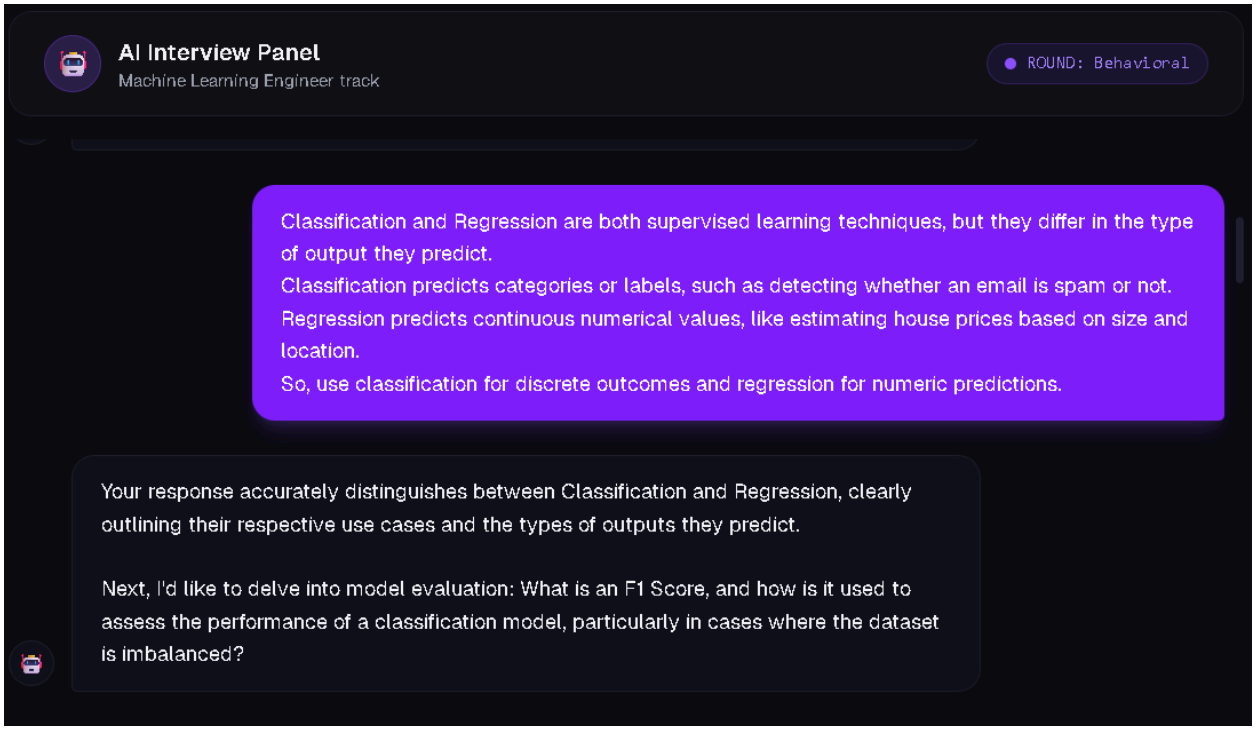


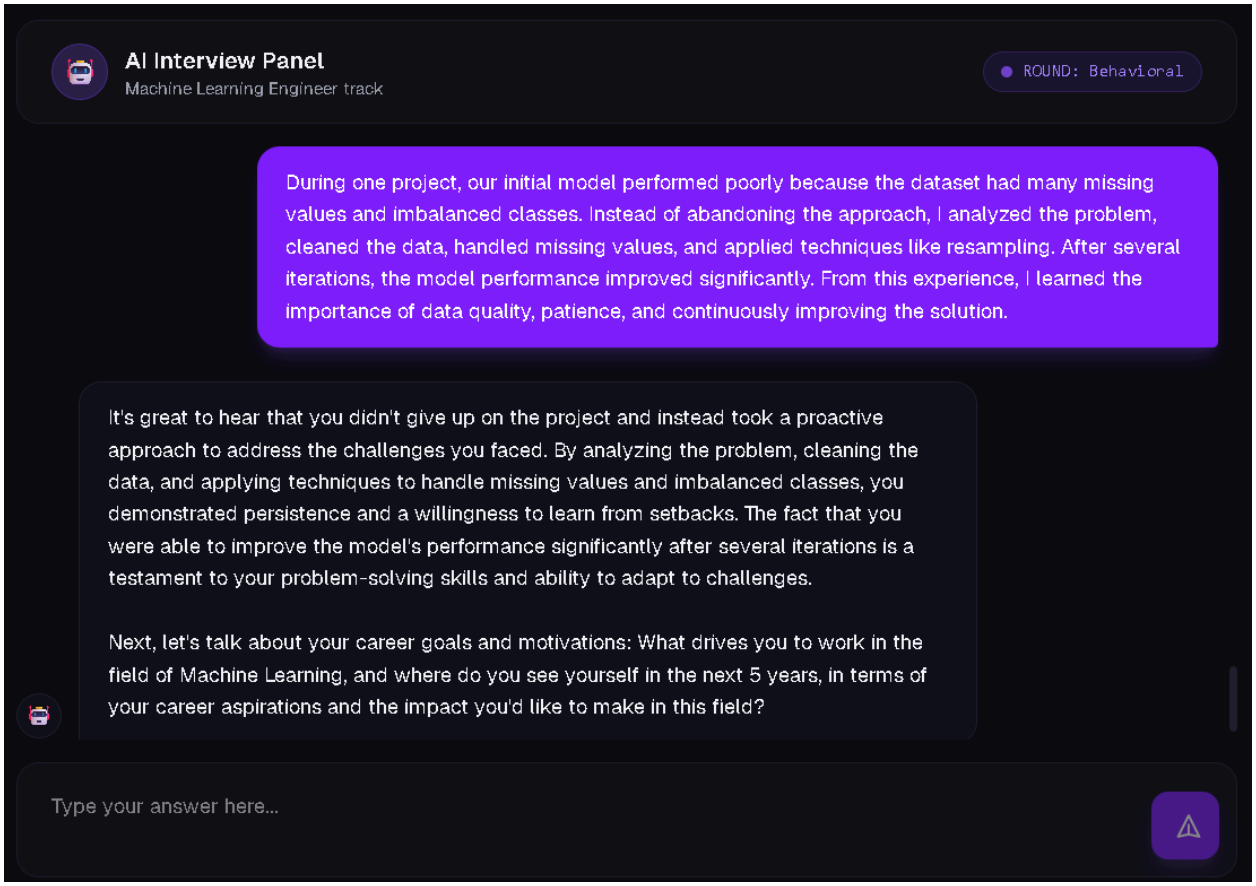

Skills
TECHNICAL
AI
C
Html
GO
Python
SQL
Node
Less
C++
Express
React
API
Javascript
Java
Computer Vision
R
CSS
GIT
Data Science
Google Cloud
Node.Js
Mongodb
Firebase
Machine Learning
TOOLS
Postman
Github
SOFT SKILLS
Collaboration
Problem Solving
Communication
Leadership
Experience
Intelligence And Data Science Currently building: Peer – A platform for online degree students to connect
July 2027
B.M.S. College of Engineering , Bengaluru, and engage in peer-to-peer mentorship.
India|Expected in July 2027
Built: Campus Whisper – An anonymous platform for college students to
freely express and discuss campus life.
AI
C
GO
Python
C++
Express
R
GIT
Data Science
Mongodb


Education
Aspiring full-stack developer passionate about building student-focused web
ms. in Engineering in Artificial
Continue to Onboarding →


1
When starting a project in a technology stack you have never used before, what is your typical first step? *
A Read official documentation and architecture guides first
B Build a small prototype to experiment with the technology
C Watch tutorials or study existing open-source projects
D Ask experienced teammates for recommended best practices


You're all set!
Your engineering profile has been analyzed
Your Engineering Traits
Research Efficiency 90%
Accountability 90%
Process Maturity 90%
Learning Agility 80%
Code Quality Focus 80%
Incident Management 70%
Self Awareness 70%
Security Responsibility 70%
Inclusive Design 70%
Architecture Thinking 60%


AI Interview Panel
Machine Learning Engineer track
ROUND: Behavioral
Classification and Regression are both supervised learning techniques, but they differ in the type of output they predict.
Classification predicts categories or labels, such as detecting whether an email is spam or not.
Regression predicts continuous numerical values, like estimating house prices based on size and location.
So, use classification for discrete outcomes and regression for numeric predictions.
Your response accurately distinguishes between Classification and Regression, clearly outlining their respective use cases and the types of outputs they predict.
Next, I'd like to delve into model evaluation: What is an F1 Score, and how is it used to assess the performance of a classification model, particularly in cases where the dataset is imbalanced?


AI Interview Panel
Machine Learning Engineer track
ROUND: Behavioral
During one project, our initial model performed poorly because the dataset had many missing values and imbalanced classes. Instead of abandoning the approach, I analyzed the problem, cleaned the data, handled missing values, and applied techniques like resampling. After several iterations, the model performance improved significantly. From this experience, I learned the importance of data quality, patience, and continuously improving the solution.
It's great to hear that you didn't give up on the project and instead took a proactive approach to address the challenges you faced. By analyzing the problem, cleaning the data, and applying techniques to handle missing values and imbalanced classes, you demonstrated persistence and a willingness to learn from setbacks. The fact that you were able to improve the model's performance significantly after several iterations is a testament to your problem-solving skills and ability to adapt to challenges.
Next, let's talk about your career goals and motivations: What drives you to work in the field of Machine Learning, and where do you see yourself in the next 5 years, in terms of your career aspirations and the impact you'd like to make in this field?
Type your answer here...

Interview Analysis Report
Our Cognitive Analyst has reviewed your transcript. Here is your evaluation.
OVERALL COMMUNICATION SCORE
8 / 10
"The candidate demonstrated a strong technical foundation in Machine Learning, providing clear and concise answers to most questions. However, they occasionally lacked detailed explanations of their thought process and sounded rehearsed in their responses. The candidate showed good collaboration and problem-solving skills in the scenario-based questions, but could improve by asking more clarifying questions and acknowledging potential uncertainties in their answers."

Skipping clarification

LOW SEVERITY

"The candidate rarely asked clarifying questions, but the interviewer's questions were generally clear and well-defined, minimizing the need for clarification."

Encourage the candidate to ask clarifying questions when needed to ensure understanding of the problem or question.

Weak explanation structure

LOW SEVERITY

"The candidate's explanations were generally well-organized, but occasionally lacked a clear beginning, middle, and end, such as when discussing the Bias-Variance trade-off."

Encourage the candidate to use a more structured approach to explaining complex concepts.

Overconfidence

LOW SEVERITY

"The candidate demonstrated a good understanding of Machine Learning concepts, but occasionally stated their answers with absolute certainty, without acknowledging potential uncertainties or trade-offs."

Encourage the candidate to acknowledge potential uncertainties or trade-offs when providing answers.

Detected Cognitive Patterns

Answering without reasoning

LOW SEVERITY

"The candidate often provided clear and concise definitions of Machine Learning concepts, but occasionally lacked detailed explanations of their thought process, such as when explaining the difference between Classification and Regression."

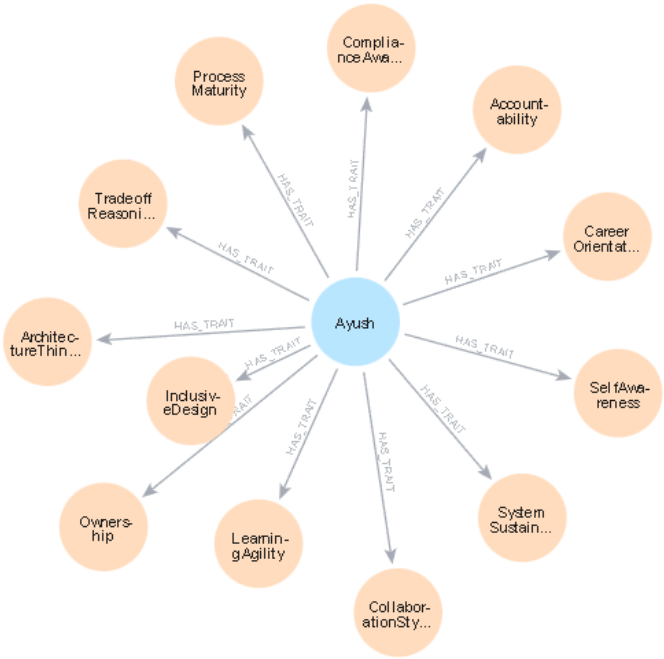
Encourage the candidate to elaborate on their thought process behind the answers.

Memorized responses

MEDIUM SEVERITY

"The candidate's responses to technical questions, such as the explanation of the F1 Score or the types of Kernels used in SVM, sounded well-rehearsed and lacked adaptation to the specific context of the question."

Ask follow-up questions that require the candidate to apply their knowledge in a more nuanced or context-dependent way.



Future Enhancements

1. **Voice-Based Interview Simulation**
   Future versions of the system can integrate speech recognition and voice interaction to simulate real interview environments. Candidates would be able to answer questions verbally, allowing the system to analyze speaking clarity, confidence, and hesitation patterns in addition to the content of the response.

2. **Advanced Knowledge Graph Reasoning**
   The knowledge graph can be enhanced with deeper concept dependency modeling and automated reasoning capabilities. This would allow the system to identify root conceptual weaknesses more accurately and generate more targeted learning paths for candidates.

3. **Personalized Learning Recommendations**
   The system can be extended to recommend learning resources such as tutorials, videos, and practice exercises based on the candidate's weak areas. This would help users improve specific concepts before attempting another interview simulation.

4. **Company-Specific Interview Preparation**
   Future enhancements could include datasets of interview questions and evaluation styles from different companies. The system could then simulate interviews tailored to specific organizations, helping candidates prepare for company-specific hiring processes.